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global_data={
  "attribute_1": "total_cost",
  "attribute_2": "material_continuity",
  "attribute_3": "current_rating",
  "attribute_4": "run_length",
  "attribute_5": "voltage_class",

  "direction_attribute_1": "negative",
  "direction_attribute_2": "positive",
  "direction_attribute_3": "positive",
  "direction_attribute_4": "positive",
  "direction_attribute_5": "positive",

  "attribute_1_min": 300,
  "attribute_1_max": 200000,
  "attribute_2_min": 0,
  "attribute_2_max": 1,
  "attribute_3_min": 100,
  "attribute_3_max": 10000,
  "attribute_4_min": 3,
  "attribute_4_max": 500,
  "attribute_5_min": 200,
  "attribute_5_max": 40000,

  "copper_price_per_ton": 9500,
  "aluminum_price_per_ton": 2500,
  "copper_price_per_kg": 9.5,
  "aluminum_price_per_kg": 2.5
}
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regional_data=[
  {
    "region": "United States",
    "copper_product_market_share": 0.72,
    "aluminum_product_market_share": 0.28,
    "weight_attribute_1": 0.25,
    "weight_attribute_2": 0.22,
    "weight_attribute_3": 0.30,
    "weight_attribute_4": 0.13,
    "weight_attribute_5": 0.10
  },
  {
    "region": "China",
    "copper_product_market_share": 0.58,
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    "aluminum_product_market_share": 0.42,
    "weight_attribute_1": 0.38,
    "weight_attribute_2": 0.08,
    "weight_attribute_3": 0.28,
    "weight_attribute_4": 0.18,
    "weight_attribute_5": 0.08
  },
  {
    "region": "India",
    "copper_product_market_share": 0.48,
    "aluminum_product_market_share": 0.52,
    "weight_attribute_1": 0.38,
    "weight_attribute_2": 0.14,
    "weight_attribute_3": 0.23,
    "weight_attribute_4": 0.13,
    "weight_attribute_5": 0.12
  },
  {
    "region": "Europe",
    "copper_product_market_share": 0.67,
    "aluminum_product_market_share": 0.33,
    "weight_attribute_1": 0.20,
    "weight_attribute_2": 0.22,
    "weight_attribute_3": 0.28,
    "weight_attribute_4": 0.14,
    "weight_attribute_5": 0.16
  }
]

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product_data=[
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "Feeder Busway, Cu, 3P+N, 2000A, 480V LV, 30m",
    "copper_kg": 860,
    "aluminum_kg": 0,
    "total_mass_kg": 1290,
    "non_material_cost_per_unit": 7000,
    "attribute_2_value": 0.85,
    "attribute_3_value": 2000,
    "attribute_4_value": 30,
    "attribute_5_value": 480
  },

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{
  "region": "United States",
  "dominant material": "al",
  "product": "Feeder Busway, Al, 3P+N, 2000A, 480V LV, 30m",
  "copper_kg": 0,
  "aluminum_kg": 420,
  "total_mass_kg": 870,
  "non_material_cost_per_unit": 7000,
  "attribute_2_value": 0.15,
  "attribute_3_value": 2000,
  "attribute_4_value": 30,
  "attribute_5_value": 480
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "Feeder Busway, Cu, 3P+N, 2000A, 380V LV, 30m",
  "copper_kg": 860,
  "aluminum_kg": 0,
  "total_mass_kg": 1290,
  "non_material_cost_per_unit": 2500,
  "attribute_2_value": 0.45,
  "attribute_3_value": 2000,
  "attribute_4_value": 30,
  "attribute_5_value": 380
},
{
  "region": "China",
  "dominant material": "al",
  "product": "Feeder Busway, Al, 3P+N, 2000A, 380V LV, 30m",
  "copper_kg": 0,
  "aluminum_kg": 420,
  "total_mass_kg": 870,
  "non_material_cost_per_unit": 2500,
  "attribute_2_value": 0.55,
  "attribute_3_value": 2000,
  "attribute_4_value": 30,
  "attribute_5_value": 380
},
{
  "region": "India",
  "dominant material": "cu",
  "product": "Feeder Busway, Cu, 3P+N, 2000A, 415V LV, 30m",
  "copper_kg": 860,
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"aluminum_kg": 0,
"total_mass_kg": 1290,
"non_material_cost_per_unit": 2250,
"attribute_2_value": 0.40,
"attribute_3_value": 2000,
"attribute_4_value": 30,
"attribute_5_value": 415
},
{
  "region": "India",
  "dominant material": "al",
  "product": "Feeder Busway, Al, 3P+N, 2000A, 415V LV, 30m",
  "copper_kg": 0,
  "aluminum_kg": 420,
  "total_mass_kg": 870,
  "non_material_cost_per_unit": 2250,
  "attribute_2_value": 0.60,
  "attribute_3_value": 2000,
  "attribute_4_value": 30,
  "attribute_5_value": 415
},
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "Feeder Busway, Cu, 3P+N, 2000A, 400V LV, 30m",
  "copper_kg": 860,
  "aluminum_kg": 0,
  "total_mass_kg": 1290,
  "non_material_cost_per_unit": 6500,
  "attribute_2_value": 0.80,
  "attribute_3_value": 2000,
  "attribute_4_value": 30,
  "attribute_5_value": 400
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "Feeder Busway, Al, 3P+N, 2000A, 400V LV, 30m",
  "copper_kg": 0,
  "aluminum_kg": 420,
  "total_mass_kg": 870,
  "non_material_cost_per_unit": 6500,
  "attribute_2_value": 0.20,
  "attribute_3_value": 2000,
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    "attribute_4_value": 30,  
    "attribute_5_value": 400  
  }  
]
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